Assignment 3

① 1

1 2

1 2 3

1 2 3 4

1 2 3

1 2

1

```
for( int i=1; i≤n; i++){
    for( int j=1; j≤n-i; j++){
        sop("__");
    }
    for( int p=1; p≤i; p++){
        sop(p+"__");
    }
    sopln();
}
for( int i=n-1; i≥1; i--){
    for( int j=1; j≤n-i; j++){
        sop("  ");
    }
    for (int p=1; p≤i; p++){
        sop(p+"__");
    }
    sopln();
}
```

Dry run :-

i=1 ; 1≤4

j=1 , 1≤4-1 1≤3 .

T → —

j++ j=2; 2≤4-1 2≤3

T → —

j++ j=3; 3≤4-1 3≤3

T → —

j++ j=4 , 4≤4-1 4≤3

False

$P = 1;\ 1 \le 1$

$T \rightarrow 1$

$P++$

$P = 2;\ 2 \le 4$

$j = 1,\ j \le 4-2 \quad 1 \le 2$

$i++$ $\quad T \rightarrow -$

$j++ \quad j = 2,\ j \le 4-2 \quad 2 \le 2$

$T \rightarrow -$

$P = 1;\ 1 \le 2$

$T \rightarrow 1$

$P++ \quad P = 2,\ 2 \le 2$

$T \rightarrow 2$

$i++$

$\boxed{i = 3},\ 3 \le 4$

$j = 1;\ j \le 4-3 \quad 1 \le 1$

$T \rightarrow -$

$P = 1;\ 1 \le 3$

$T \rightarrow 1$

$P++ \quad P = 2;\ 2 \le 3$

$T \rightarrow 2$

$P++ \quad P = 3,\ 3 \le 3$

$T \rightarrow 3$

$i++$

$\boxed{i = 4},\ 4 \le 4$

$j = 1,\ j \le 4-4 \quad 1 \le 0$

false

$P = 1,\ 1 \le 4$

$T \rightarrow 1$

$P++ \quad P = 2,\ 2 \le 4$

$T \rightarrow 2$

$P++ \quad P = 3,\ 3 \le 4$

$T \rightarrow 3$

$P++ \quad P = 4,\ 4 \le 4$

$T \rightarrow 4$

(2)

```
1
1 2                     1
1 2 3                 2 1
1 2 3 4 4           3 2 1
1 2 3 4 4           3 2 1
1 2 3               3 2 1
1 2                   2 1
1                       2 1
```

→

```
for( int i=1; i<=n; i++){
    for(int j=1; j<=i, j++){
        sop (j);
    }
    for(int p=1; p<=2x(n-i); p++){
        Sop(" ");
    }
    for (int j=i; j>=1; j--){
        sop (j);
    }
    Sopln();
}
for( int i=n; i>=1; i--){
    for(int j=1; j<=i; j++){
        sop(j);
    }
    for(int p=1; p<=2x(n-i); p++){
        sop(" ");
    }
    for(int j=i; j>=1; j--){
        sop(j);
    }
    sop();
}
}
```

(2)

```
1
1 2                     1
1 2 3                 2 1
1 2 3 4 4           3 2 1
1 2 3 4 4           3 2 1
1 2 3               3 2 1
1 2                   2 1
1                       2 1
```

→

```
for( int i=1; i<=n; i++){
    for(int j=1; j<=i , j++){
        sop (j);
    }
    for(int p=1; p<=2x(n-i); p++){
        Sop(" ");
    }
    for (int j=i; j>=1; j--){
        sop (j);
    }
    Sopln();
}
for( int i=n; i>=1; i--){
    for(int j=1; j<=i; j++){
        sop(j);
    }
    for(int p=1; p<=2x(n-i); p++){
        sop(" ");
    }
    for(int j=i; j>=1; j--){
        sop(j);
    }
    sop();
}
}
```

int i=n-1; i≥1; i--

$\boxed{i=3}$ $3 \geq 1$

j=1, j ≤ 4-3 $1 \leq 1$

T → –

j++ j=2; $2 \leq 1$

false

p=1; $1 \leq 3$

T → 1

p++ p=2, $2 \leq 3$

T → 2

p++ p=3, $3 \leq 3$

T → 3

i-- $\boxed{i=2}$ $2 \geq 1$

j=1; j<4-2 $1 \leq 2$

T → –

j++ j=2, $2 \leq 2$

T → –

p=1; $1 \leq 2$

T → 1

p++ p=2, $2 \leq 2$

T → 2

i-- $\boxed{i=1}$; $1 \geq 1$

j=1 $1 \leq 4-1$ $1 \leq 3$

T ↳ –

j++ j=2; $2 \leq 4-2$, $2 \leq 2$

T → –

j++ j=3; $1 \leq 4-3$ $3 \leq 2$

false

p=1; $1 \leq 1$

T ↳ 1

j-- j=1, 1≥1

T → 1

i++

$\boxed{i=3}$ 3≤4

j=1, 1≥3

T → 1

j++ j=2; 2≤3

T → 2

j++ j=3, 3≤3

T → 3

p=1, p≤2×(4-3) 1≤2

T → —

p++ p=2, 2≤2

T → —

j=3, 3≥1

T → 3

j-- j=2, 2≥1

T → 2

j-- j=1; 1≥1

T → 1

i++

$\boxed{i=4}$ 4≤4

j=1, 1≤4

T → 1

j++ j=2, 2≤4

T → 2

j++ j=3, 3≤4

T → 3

j++ j=4, 4≤4

T → 4

p=1, p≤2×(4-4) 1≤0

false

j=4; 4≥1

T → 4

Dry run :-

$\boxed{i=1}$, $i \le 4$

$j=1$, $j \le 1$

$T \rightarrow 1$

$j++$ $j=2$, $2 \le 1$

false

$p=1$; $p \le 2 \times (4-1)$ $1 \le 2 \times 3$ $1 \le 6$

$T \rightarrow$ —

$p++$ $p=2$, $2 \le 6$

$T \rightarrow$ —

$p++$ $p=3$, $3 \le 6$

$T \rightarrow$ —

$p++$ $p=4$, $4 \le 6$

$T \rightarrow$ —

$p++$ $p=5$, $5 \le 6$

$T \rightarrow$ —

$p++$ $p=6$, $6 \le 6$

$T \rightarrow$ —

$j=1$, $1 \ge 1$

$T \rightarrow 1$

$i++$

$\boxed{i=2}$, $2 \le 4$

$j=1$, $1 \le 2$

$T \rightarrow 1$

$j++$ $j=2$, $2 \le 2$

$T \rightarrow 2$

$p=1$, $p \le 2 \times (4-2)$ $p \le 2(2)$ $2 \le 6$

$T \rightarrow$ —

$p++$ $p=2$, $2 \le 4$

$T \rightarrow$ —

$p++$ $p=4$, $4 \le 4$

$T \rightarrow$ —

$j=2$, $2 \ge 1$

$T \rightarrow 2$

j-- j=2, 2≥1
T → 2
j-- j=1, 1≥1
T → 1
i-- i=2, 2≥1
j=1, 1≤2
T → 1
j++ j=2, 2≤2
T → 2
j=2, 2≥1
T → 2
j-- j=1, 1≥1
T → 1
i-- i=1, 1≥1
j=1, 1≤1
T → 1
j=1, 1≥1
T → 1

(9)

```
1        1
  2    2
    3
  4    4
5        5
```

→

```
for(int i=1; i≤n; i++){
    for(int j=1; j≤n; j++){
        if ((i==j)||(i+j==n+1)){
            sop(i+" ");
        }
        else{
            sop(" ");
        }
    }
    sopln();
}
```

j-- j=3, 3≥1
T → 3
j-- j=2, 2≥1
T → 2
j-- j=1, 1≥1
T → 1

$\boxed{i=4}$ 4≥1
j=1, 1≤4
T → 1
j++ j=2, 2≤4
T → 2
j++ j=3, 3≤4
T → 3
j++ j=4, 4≤4
T — 4
j=4, 4≥1
T→4
j-- j=3, 3≥1
T → 3
j-- j=2, 2≥1
T → 2
j-- j=1 1≥1
T → 1

$\boxed{i=3}$ 3≥1
j=1 ; 1≤3
T → 1
j++ j=2, 2≤3
T → 2
j++ j=3, 3≤3
T → 3
j=3, 3≥1
T → 3

Dry run :-

$\boxed{i=1}$ $1 \le 5 \rightarrow T$

$j=1,\ 1 \le 5$

$(1==1) \rightarrow T$

$\hookrightarrow 1$

j++ $j=2,\ 2 \le 5 \rightarrow$ False

$T \rightarrow (1==2) \,||\, (1+2==5+1)$

$\hookrightarrow -$

j++ $j=3,\ 3 \le 5$

$(1==3) \,||\, (1+3==6) \rightarrow T$

$\hookrightarrow -$

j++ $j=4,\ 4 \le 5 \rightarrow$ False

$(1==4) \,||\, (4+1==6)$

$\rightarrow -$

j++ $j=5,\ 5 \le 5 \rightarrow T$

$(1==5) \,||\, (5+1==6)$

$\rightarrow 1$

i++

$\boxed{i=2}$ $2 \le 5$

$j=1,\ 1 \le 5$ False

$(2==1) \,||\, (2+1==6)$

$\rightarrow -$

j++ $j=2,\ 2 \le 5 \rightarrow T$

$(2==2)$

$T \rightarrow 2$

j++ $j=3,\ 3 \le 5$ False

$(2==3) \,||\, (2+3==6)$

$\rightarrow -$

j++ $j=4,\ 4 \le 5$

$(2==4) \,||\, (2+4==6) \rightarrow T$

$\rightarrow 2$

j++ $j=5,\ 5 \le 5$

$(2==5) \,||\, (2+5==6) \rightarrow T$

$\rightarrow -$

(7) Dry run :-

int num = 1;

i = 1 ; 1 ≤ 4

p = 1 ; 1 ≤ 4-1 1 ≤ 3

T → —

p++ p = 2, 2 ≤ 3

T → —

p++ p = 3, 3 ≤ 3

T → —

j = 1, 1 ≤ 1

(1 % 2 == 1)

T → 1

j++

j = 2, 2 ≤ 1 → false

i++

$\boxed{i = 2}$ 2 ≤ 4

T → p = 1, 1 ≤ 4-2, 1 ≤ 2

T → —

p++ p = 2, 2 ≤ 2

T → —

j = 1, 1 ≤ 2

(1 % 2 == 1)

T → 1

j++ j = 2, 2 < 2

(2 % 2 == 1)

→ 0

i++

$\boxed{i = 3}$ 3 ≤ 4

p = 1, p ≤ 4-3, 1 ≤ 1

T → —

j = 1, 1 ≤ 3

(1 % 2 == 1)

T → 1

i++ $\boxed{i=4}$ $4 \le 5$

$j = 1, 1 \le 5$

$(4 == 1 \;||\; 4+1 == 6)$

$\rightarrow$ —

j++ $j = 2; 2 \le 5$

$(4 == 2 \;||\; 4+2 == 6)$

$T \rightarrow 4$

j++ $j = 3, 3 \le 5$

$(4 == 3 \;||\; 4+3 == 6) \rightarrow F$

$\hookrightarrow$ —

j++ $j = 4, 4 \le 5$

$(4 == 4 \;||\; 4+4 == 6)$

$\hookrightarrow 4$

j++ $j = 5, 5 \le 5$

$(4 == 5 \;||\; 4+5 == 6) \rightarrow F$

$\hookrightarrow$ —

i++ $\boxed{i=5}$, $5 \le 5$

$j = 1, 1 \le 5$

$(5 == 1) \;||\; (5+1 == 6)$

$T \rightarrow 5$

j++, $2 \le 5$

$(5 == 2) \;||\; (5+2 == 6) \rightarrow F$

$\hookrightarrow$ —

j++, $3 \le 5$

$(5 == 3) \;||\; (5+3 == 6) \rightarrow F$

$\rightarrow$ —

j++ ; $4 \le 5$

$(5 == 4) \;||\; (5+4 == 6) \rightarrow F$

$\rightarrow$ —

j++ $5 \le 5$

$(5 == 5) \;||\; (5+5 == 6)$

$\rightarrow 5$

j++ j=3, 3 ≤ 3
(3 % 2 == 1)
sop → 1

i-- i=2, 2 ≥ 1
p=1; 1 ≤ 4-2, 1 ≤ 2
sop → —

p++ p=2; 2 ≤ 2
~~1~~ → —

j=1; 1 ≤ 2
(1 % 2 == 1)
T → 1

j++ j=2; 2 ≤ 2
(2 % 2 == 1)
T → 0

i-- (i=1) 1 ≥ 1
p=1; 1 ≤ 4-1 1 ≤ 3
T → —

p++ p=2; 2 ≤ 4-1 2 ≤ 3
T → —

p++ p=3, 3 ≤ 4-1 3 ≤ 3
T → —

j=1; 1 ≤ 1
1 % 2 == 1
T → 1

(4)

3 3 3 3 3
3 2 2 2 3
3 2 1 2 3
3 2 2 2 3
3 3 3 3 3

```
int size = 2*n-1;
for( int i=1; i≤size; i++){
    for( int j=1; j≤size; j++){
```

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j++

j = 2; 2 ≤ 3

(2 % 2 == 1)

→ 0

j++ j = 3; 3 ≤ 3

(3 % 2 == 1)

→ 1

i++

$\boxed{i = 4}$, 4 ≤ 4

p = 1, I ≤ 4 - 4 I ≤ 0

False

j = 1, 1 ≤ 4

(1 % 2 == 1)

→ 1

j++ j = 2, 2 ≤ 4

(2 % 2 == 1) → T

T → 0

j++ j = 3, 3 ≤ 4

(3 % 2 == 1) → T

T → 1

j++ j = 4; 4 ≤ 4

(4 % 2 == 1)

→ 0

i = 4 - 1 $\boxed{i = 3}$ 3 ≤ 1

p = 1, 1 ≤ 4 - 3, i ≤ 1

sop → —

j = 1 ; 1 ≤ 3

(1 % 2 == 1)

sop → 1

j++ . j = 2, 2 ≤ 3

(2 % 2 == 1)

→ 0

```
        1
      2   3
    4   5   6
  7   8   9   10
11   12   13
  14   15
     16
int num=1;
for(int i=1; i≤n; i++){
  for(int p=1; p≤n-i; p++){
      sop(" ");
  }
  for(int j=1; j≤i; j++){
      sop(num+" ");
      sopln();
  }
  for(int i=n-1; i≥1; i--){
  for(int p=1; p≤n-i; p++){
      sop(" ");
  }
  for(int j=1; j≤i; j++){
      sop(num+" ");
  }
      sopln();
}}}
```

Dry run :-

int num=1;

i=1; n=4, 1≤4

p=1; 1 ≤ 4-1 1≤3

T → —

p++ p=2; p 2≤3

T → —

num=10

j++ j=4; 4≤4

T → 10

num=11

i=i-1 i=3; 3≥1

p=1; 1≤4-3, 1≤1

T → –

j=1 ; 1≤3

T → 11

num=12

j++ j=2, 2≤3

T → 12

num=13

j++ j=3; 3≤3

T → 13

num=14

i--

i=2 ; 2≥1

p=1 ; 1≤4-2 1≤2

T → –

p++ p=2 ; 2≤2

T → –

j=1; 1≤2

T → 14

num=15

j++ j=2; 2≤2

T → 15

j++ j=2 2≤2

T → 15

num=16

i-- $\boxed{i=1}$ 1≤1

T → p=1 ; 1≤3-1 1≤2

T → –

p++ p=3; 3≤3
T → –
j=1, 1≤1
T → 1
i++ num=2
i=2; 2≤4
p=1; 1≤4-2 1≤2
T → –
p++ p=2, 2≤2
T → –
j=1; 1≤2
T → 2
p++ num=3
j++ j=2; 2≤2
T → 3
i++ num=4
p=1; 1≤4-3 1≤1
T → –
j=1; 1≤3
T → 4
i++ num=5
j++ j=3; 3≤3
T → 6
num=7
i= 4≤4
p=1, 1≤4-4 1≤0
False
j=1, 1≤4
T → 7
num=8
j++ j=2, 2≤4
T → 8
num=9
j++ j=3, 3≤4
T → 9

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p++ p=2 ; 2 ≤ 2
T → –
j = 1 ; 1 ≤ 1
T → 16